Operations with Matrices

- Matrix:

$$A=[a_{ij}]=\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m\times n} \in M_{m\times n}$$

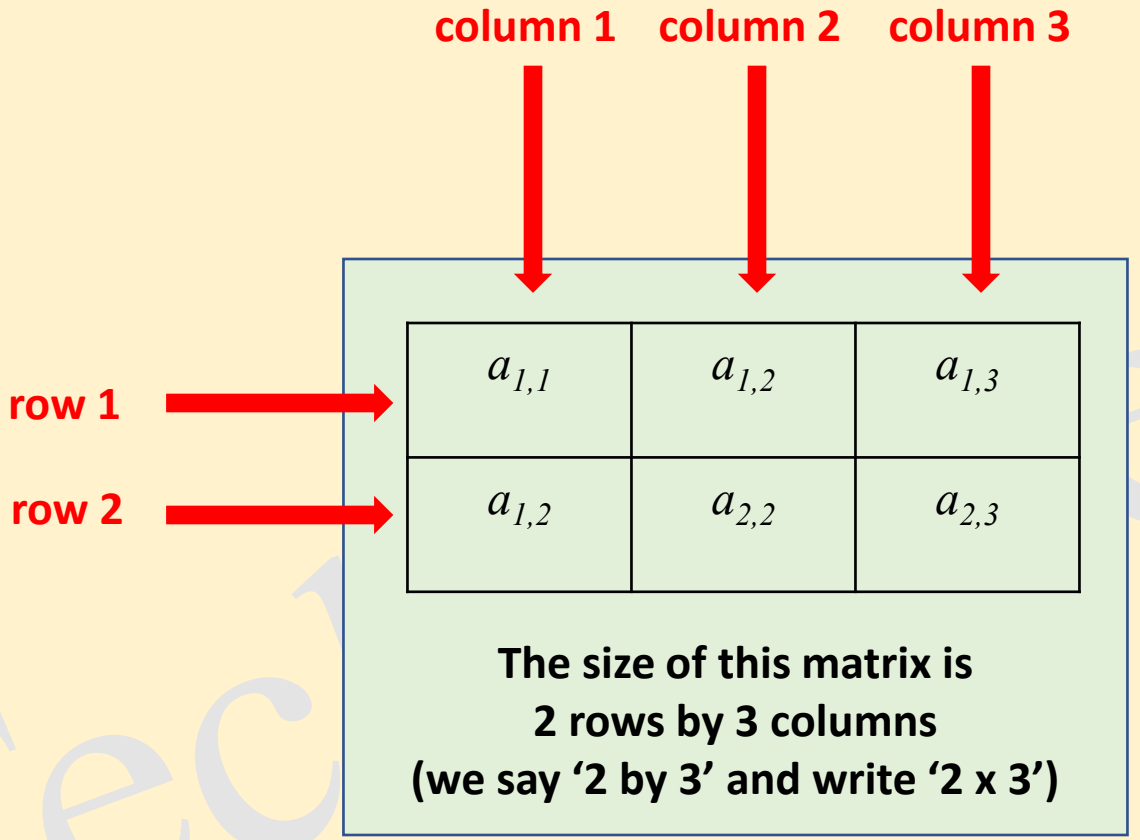
(i,j)-th entry: a_{ij}

row: m

column: n

size: $m\times n$

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column 1
column 2
column 3
row 1
row 2
$a_{1,1}$
$a_{1,2}$
$a_{1,3}$
$a_{1,2}$
$a_{2,2}$
$a_{2,3}$
The size of this matrix is
2 rows by 3 columns
(we say '2 by 3' and write '2 x 3')

- Row matrix

$$r = [a_{11}\ a_{12}\ \ \dots a_{1n}]_{1\times n}$$

- Column matrix

$$c = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}_{m\times 1}$$

- Square matrix: $m = n$

$$B_{n\times n} = \begin{bmatrix} a_{11} & \cdots & a_{n1} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}_{n\times n}$$

e.g., $B = \begin{bmatrix} 4 & 5 \\ 3 & 2 \end{bmatrix}$

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- *i*-th row vector of A

$$r_i = \begin{bmatrix} a_{i1} & a_{i2} & \cdots & a_{in} \end{bmatrix}$$ row matrix

- *j*-th column vector A

$$c_j = \begin{bmatrix} c_{1j} \\ c_{2j} \\ \vdots \\ c_{mj} \end{bmatrix}$$ column matrix

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- Ex:

$$A=\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\end{bmatrix}=\begin{bmatrix}r_1\\ r_2\end{bmatrix}$$

$$\Rightarrow \quad r_1=\begin{bmatrix}1 & 2 & 3\end{bmatrix},\ r_2=\begin{bmatrix}4 & 5 & 6\end{bmatrix}$$

$$A=\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\end{bmatrix}=\begin{bmatrix}c_1 & c_2 & c_3\end{bmatrix}$$

$$\Rightarrow c_1=\begin{bmatrix}1\\ 4\end{bmatrix},\ c_2=\begin{bmatrix}2\\ 5\end{bmatrix},\ c_3=\begin{bmatrix}3\\ 6\end{bmatrix}$$

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- Diagonal matrix:

$$A = diag(d_1, d_2, \cdots, d_n) = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix} \in M_{n\times n}$$

- Trace:

If $A = [a_{ij}]_{n\times n}$

Then $Tr(A) = a_{11} + a_{22} + \cdots + a_{nn}$

- Scalar Matrix:

If in a diagonal matrix all diagonal elements are same scalar.

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- Identity Matrix:

If all diagonal elements of scalar matrix are 1

- Triangular matrix:

(i) A square matrix is said to be an upper triangular if all the elements below the diagonal are zero.

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 5 & 6 \\ 0 & 0 & 7 \end{bmatrix}$$

(ii) A square matrix is said to be a lower triangular if all the elements above the diagonal are zero.

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 3 & 5 & 0 \\ 4 & 6 & 7 \end{bmatrix}$$

(iii) A square matrix is said to be a triangular matrix if it is upper or lower triangular matrix.

(iv) A square matrix is said to be a strictly triangular matrix if all diagonal elements are zero.

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- Equal matrix:

If $A=[a_{ij}]_{m\times n}$, $B=[b_{ij}]_{m\times n}$

Then $A=B$ if and only if $a_{ij}=b_{ij}\ \forall 1\le i\le m,\ 1\le j\le n$

- Ex 1: (Equal matrix)

$$A=\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix} \qquad B=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$$

If $A=B$

Then $a=1,\ b=2,\ c=3,\ d=4$

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2. Properties of Matrix Operations

- Three basic matrix operators:

  (1) matrix addition

  (2) scalar multiplication

  (3) matrix multiplication

- Zero matrix: $0_{m\times n}$

- Identity matrix of order n: I_n

$$I_n = \begin{bmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{bmatrix}$$

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- Properties of zero matrices:

If $A \in M_{m\times n}$, c : scalar

Then (1) $A + 0_{m\times n} = A$

(2) $A + (-A) = 0_{m\times n}$

(3) $cA = 0_{m\times n} \Rightarrow c = 0$ *or* $A = 0_{m\times n}$

- Notes:

(1) $0_{m\times n}$: **the additive identity** for the set of all $m \times n$ matrices

(2) $-A$: **the additive inverse** of A

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- Matrix addition:

If $A=[a_{ij}]_{m\times n}$, $B=[b_{ij}]_{m\times n}$

Then $A+B=[a_{ij}]_{m\times n}+[b_{ij}]_{m\times n}=[a_{ij}+b_{ij}]_{m\times n}$

- Ex 2: (Matrix addition)

$$\begin{bmatrix}-1 & 2\\ 0 & 1\end{bmatrix}+\begin{bmatrix}1 & 3\\ -1 & 2\end{bmatrix}=\begin{bmatrix}-1+1 & 2+3\\ 0-1 & 1+2\end{bmatrix}=\begin{bmatrix}0 & 5\\ -1 & 3\end{bmatrix}$$

$$\begin{bmatrix}1\\ -3\\ -2\end{bmatrix}+\begin{bmatrix}-1\\ 3\\ 2\end{bmatrix}=\begin{bmatrix}1-1\\ -3+3\\ -2+2\end{bmatrix}=\begin{bmatrix}0\\ 0\\ 0\end{bmatrix}$$

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- Scalar multiplication:

  If $A=[a_{ij}]_{m\times n}$, c : scalar

  Then $cA=[ca_{ij}]_{m\times n}$

- Matrix subtraction:

  $A-B=A+(-1)B$

- Ex 3: (Scalar multiplication and matrix subtraction)

$$A=\begin{bmatrix}1 & 2 & 4\\ -3 & 0 & -1\\ 2 & 1 & 2\end{bmatrix} \qquad B=\begin{bmatrix}2 & 0 & 0\\ 1 & -4 & 3\\ -1 & 3 & 2\end{bmatrix}$$

Find $(a)\ 3A$, $(b)\ -B$, $(c)\ 3A-B$

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- Properties of matrix addition and scalar multiplication:

If $A, B, C \in M_{m\times n}$, c, d : scalar

Then (1) $A+B = B+A$

(2) $A+(B+C)=(A+B)+C$

(3) $(cd)A = c(dA)$

(4) $1A = A$

(5) $c(A+B) = cA + cB$

(6) $(c+d)A = cA + dA$

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- Matrix multiplication:

https://parnabdas.github.io/matrix_multiplication/?utm_source=chatgpt.com

If $A = [a_{ij}]_{m\times n}$, $B = [b_{ij}]_{n\times p}$

Then $AB = [a_{ij}]_{m\times n}[b_{ij}]_{n\times p} = [c_{ij}]_{m\times p}$

Size of AB

where $c_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} b_{11} & \cdots & b_{1j} & \cdots & b_{1n} \\ b_{21} & \vdots & b_{2j} & \cdots & b_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ b_{n1} & \cdots & b_{nj} & \cdots & b_{nn} \end{bmatrix} = \begin{bmatrix} & & & & \\ c_{i1} & c_{i2} & \cdots & c_{ij} & \cdots & c_{in} \\ & & & & \end{bmatrix}$$

- Notes: (1) $A+B = B+A$, (2) $AB \neq BA$

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- Ex 4: (Find AB)

$$A=\begin{bmatrix}-1 & 3\\ 4 & -2\\ 5 & 0\end{bmatrix} \qquad B=\begin{bmatrix}-3 & 2\\ -4 & 1\end{bmatrix}$$

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- Real number:

$ab = ba$ (Commutative law for multiplication)

- Matrix:

$$\underset{m\times n}{A}\underset{n\times p}{B} \neq \underset{n\times p}{B}\underset{m\times n}{A}$$

Three situations:

(1) If $m \neq p$, then AB is defined, BA is undefined.

(2) If $m = p, m \neq n$, then $AB \in M_{m\times m}$, $BA \in M_{n\times n}$ (Sizes are not the same)

(3) If $m = p = n$, then $AB \in M_{m\times m}$, $BA \in M_{m\times m}$

(Sizes are the same, but matrices are not equal)

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- Ex 4:

Sow that AB and BA are not equal for the matrices.

$$A=\begin{bmatrix}1 & 3\\ 2 & -1\end{bmatrix} \quad \text{and} \quad B=\begin{bmatrix}2 & -1\\ 0 & 2\end{bmatrix}$$

Sol:

$$AB=\begin{bmatrix}1 & 3\\ 2 & -1\end{bmatrix}\begin{bmatrix}2 & -1\\ 0 & 2\end{bmatrix}=\begin{bmatrix}2 & 5\\ 4 & -4\end{bmatrix}$$

$$BA=\begin{bmatrix}2 & -1\\ 0 & 2\end{bmatrix}\begin{bmatrix}1 & 3\\ 2 & -1\end{bmatrix}=\begin{bmatrix}0 & 7\\ 4 & -2\end{bmatrix}$$

- Note: $AB \neq BA$

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- Real number:

$$ac = bc,\ c \neq 0$$

$$\Rightarrow\ a = b \qquad \text{(Cancellation law)}$$

- Matrix:

$$AC = BC \qquad C \neq 0$$

(1) If C is invertible, then $A = B$

(2) If C is not invertible, then $A \neq B$ (Cancellation is not valid)

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- Ex 5: (An example in which cancellation is not valid)

  Show that $AC=BC$, *however* $A\neq B$

$$A=\begin{bmatrix}1 & 3\\ 0 & 1\end{bmatrix},\quad B=\begin{bmatrix}2 & 4\\ 2 & 3\end{bmatrix},\quad C=\begin{bmatrix}1 & -2\\ -1 & 2\end{bmatrix}$$

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- **Transpose of a matrix:**

$$\text{If } A=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \in M_{m\times n}$$

$$\text{Then } A^{T}=\begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \vdots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{bmatrix} \in M_{n\times m}$$

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- Ex 8: (Find the transpose of the following matrix)

$$(a)\quad A=\begin{bmatrix}2\\8\end{bmatrix} \qquad (b)\quad A=\begin{bmatrix}1 & 2 & 3\\4 & 5 & 6\\7 & 8 & 9\end{bmatrix} \qquad (c)\quad A=\begin{bmatrix}0 & 1\\2 & 4\\1 & -1\end{bmatrix}$$

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- Properties of transposes:

(1) $(A^T)^T = A$

(2) $(A+B)^T = A^T + B^T$

(3) $(cA)^T = c(A^T)$

(4) $(AB)^T = B^T A^T$

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- Symmetric matrix:

  A square matrix A is **symmetric** if $A = A^T$

- Skew-symmetric matrix:

  A square matrix A is **skew-symmetric** if $A^T = -A$

- Ex:

If $A = \begin{bmatrix} 1 & 2 & 3 \\ a & 4 & 5 \\ b & c & 6 \end{bmatrix}$ is symmetric, find a, b, c?

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- Ex:

If $A=\begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ b & c & 0 \end{bmatrix}$ is a skew-symmetric, find a, b, c?

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- Ex:

If $A=\begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ b & c & 0 \end{bmatrix}$ is a skew-symmetric, find a, b, c?

- Note: AA^T is symmetric

Pf: $(AA^T)^T = (A^T)^T A^T = AA^T$

$\therefore AA^T$ is symmetric

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Property 1
If A and B are two matrices and $(A + B)$ is defined then $(A + B)^T = A^T + B^T$.

Property 2
If c is a scalar then $(cA)^T = cA^T$

Property 3
If A and B are matrices of same order/ dimension then for any two scalar c and d, $(cA + dB)^T = cA^T + dB^T$.

Property 4
If for two matrices A and B, AB is defined, then $(AB)^T = B^T A^T$

Property 5
If for two matrices A and B are symmetric matrix then, $A + B$ is also a symmetric matrix.

Property 6
For any square matrix A, $A + A^T$ is a symmetric matrix and $A - A^T$ is a skew symmetric matrix.

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Q. Express the following matrix A in the form of $P + Q$, where P is a symmetric matrix and Q is a skew-symmetric matrix.

$$A = \begin{bmatrix} 4 & 5 & 1 \\ 3 & 7 & 2 \\ 1 & 6 & 8 \end{bmatrix}$$

Hint.: Use property 6.

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- Power of a square matrix:

$$(1)\ A^0 = I$$

$$(2)\ A^k = \underbrace{AA\cdots A}_{k \text{ factors}} \qquad (k > 0)$$

$$(3)\ A^r \cdot A^s = A^{r+s} \qquad r, s : \text{integers}$$

$$(A^r)^s = A^{rs}$$

$$(4)\ D = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix} \Rightarrow D^k = \begin{bmatrix} d_1^k & 0 & \cdots & 0 \\ 0 & d_2^k & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & d_n^k \end{bmatrix}$$

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1. Idempotent Matrix
A square matrix A is called **idempotent** if $A^2 = A$

2. **Nilpotent Matrix**
A square matrix A is **nilpotent** if there exists a positive integer k such that $A^k = 0$.
The smallest such k is called the **index of nilpotency**.

3. Involutory Matrix
A square matrix A is **involutory** if $A^2 = I$

4. Tripotent Matrix
A matrix A is **tripotent** if $A^3 = A$.

5. **Hermitian Matrix**
For a complex matrix A, A is **Hermitian** if $A^* = A$,
where A^* is the conjugate transpose.

6. Skew-Hermitian Matrix
A square **complex matrix** A is called a **skew-Hermitian matrix** (or anti-Hermitian matrix) if $A^* = -A$

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$$\boxed{A^2 = A \quad \text{(idempotent)}}$$

$$\boxed{A^2 = I \quad \text{(involutory)}}$$

$$\boxed{A^k = 0 \quad \text{(nilpotent)}}$$

$$\boxed{A^3 = A \quad \text{(tripotent)}}$$

A is skew-Hermitian $\Leftrightarrow$ iA is Hermitian
A is Hermitian $\Leftrightarrow$ iA is skew-Hermitian.

Matrix	**Condition**
Hermitian	$A^* = A$
Skew-Hermitian	$A^* = -A$
Symmetric	$A^T = A$
Skew-symmetric	$A^T = -A$

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2.3 The Inverse of a Matrix

- Inverse matrix:

Consider $A \in M_{n\times n}$

If there exists a matrix $B \in M_{n\times n}$ such that $AB = BA = I_n$,

Then (1) A is **invertible** (or **nonsingular**)

(2) B is **the inverse** of A

- Note:

A matrix that does not have an inverse is called **noninvertible** (or **singular**).

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- Thm 2.7: (The inverse of a matrix is unique)

  If B and C are both inverses of the matrix A, then $B = C$.

  Pf:
  $$AB = I$$
  $$C(AB) = CI$$
  $$(CA)B = C$$
  $$IB = C$$
  $$B = C$$

  Consequently, the inverse of a matrix is unique.

- Notes:

  (1) The inverse of A is denoted by A^{-1}

  (2) $AA^{-1} = A^{-1}A = I$

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- Thm 2.8： (Properties of inverse matrices)

If A is an invertible matrix, k is a positive integer, and c is a scalar not equal to zero, then

(1) A^{-1} is invertible and $(A^{-1})^{-1} = A$

(2) A^k is invertible and $(A^k)^{-1} = \underbrace{A^{-1}A^{-1}\cdots A^{-1}}_{k \text{ factors}} = (A^{-1})^k = A^{-k}$

(3) cA is invertible and $(cA)^{-1} = \frac{1}{c}A^{-1}, c \neq 0$

(4) A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$

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Important Formulas

$$\text{adj}(A) = [\text{Cofactor matrix of } A]^T$$

$$A^{-1} = \text{adj}(A) / |A|, \quad |A| \neq 0$$

If $|A| = 0$, the matrix is singular and its inverse does not exist.

1. Find inverse of

https://parnabdas.github.io/inverse_of_matrix/

$$B = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

2. Find the cofactors and inverse of

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}$$

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Properties:

1. $\det(\mathrm{A}) = \det(A^T)$

2. If two rows (or columns) are interchanged in a determinant then the sign will be changed.

3. In in an $n \times n$ matrix A if two rows are identical then $\det(A) = 0$.

4. $\det(AB) = \det(A)\det(B)$

5. A matrix A is called singular matrix if $\det(A) = 0$.

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Orthogonal Matrices:

A real $n \times n$ matrix A is said to be Orthogonal Matrix iff $AA^T = A^T A = I_n$

Property 1: If A is an orthogonal matrix, then A is nonsingular and $\det(A) = \pm 1$

Hint: $AA^T = I_n$, so $\det(AA^T) = \det(I_n) = 1$

Therefore, $\det(A)\det(A^T) = 1$, remember: $\det(A) = \det(A^T)$

Property 2:

If A is an orthogonal matrix, then $A^{-1} = A^T$

Property 3:

If A and B are orthogonal matrix of same order,
then AB is also orthogonal matrix

Unitary Matrix: For complex matrices, the corresponding concept is a **unitary matrix**: $A^* A = AA^* = I$. So, $A^{-1} = A^*$.

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Questions:

1. **Determinant Multiplication**

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 0 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 & 0 \\ 3 & 1 & 2 \\ 2 & 0 & 1 \end{bmatrix},$$

find the value of $\det(AB)$ and verify that $\det(AB) = \det(A)\det(B)$.

2. If $A = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$, show that $A^2 - 2A + I = 0$, using this find out A^{50}.

3. If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, then show that $A^2 - 4A + 5I = 0$. Now find out B such that $AB = I$.

4. For the matrices given in question 1 verify that $tr(A + B) = tr(A) + tr(B)$, $tr(AB) = tr(BA)$, $tr(A^T) = tr(A)^T$.

5. If $AB = B$ and $BA = A$, then show that A and B both are idempotent.

6. Is there exists matrices A and B such that $AB - BA = I$.

7. For which value of θ the following matrix A is orthogonal

$$A = \begin{bmatrix} cos\theta & sin\theta \\ -sin\theta & cos\theta \end{bmatrix}$$

Also find the A^{-1}.

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Questions:

8. Find the solution of the following system of equations

$$x + z = 0$$
$$3x + 4y + 5z = 2$$
$$2x + 3y + 4z = 1.$$

Is the determinant of the coefficient matrix is nonzero? If yes, then use this to solve the system of equation.

9. Is the following matrix orthogonal

$$A = \frac{1}{3}\begin{bmatrix} 1 & -2 & 2 \\ 2 & -1 & -2 \\ 2 & 2 & 1 \end{bmatrix}.$$

Using this find the solution of the following system of equation

$$x - 2y + 2z = 2$$
$$2x - y - 2z = 1$$
$$2x + 2y + z = 7.$$

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